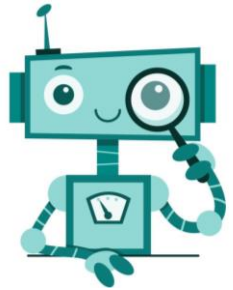
Artificial Intelligence (AI)



NCAI
Intelligent Criminology Research Lab

AGENDA

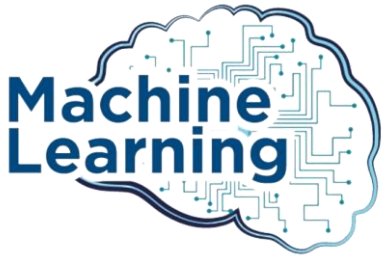
The Agenda of this presentation is:

- What is Machine Learning..?
- Supervised Machine Learning
- Unsupervised Machine Learning
- Semi Supervised Machine Learning
- Reinforcement Learning



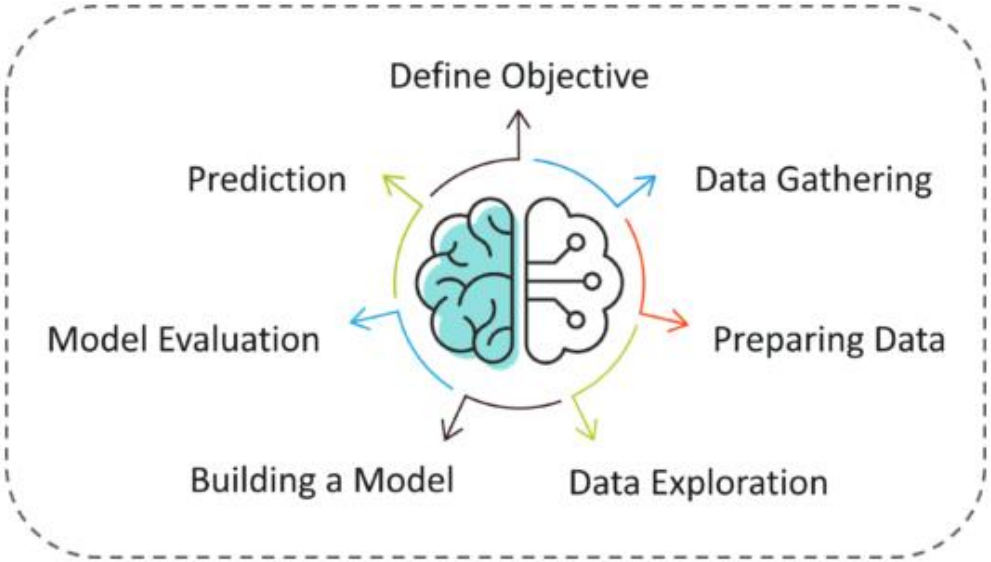
What is Machine Learning ?

- Machine Learning is making the computer learn from studying data and statistics.
- Machine Learning is a step into the direction of artificial intelligence (AI).
- Machine Learning is a program that analyses data and learns to predict the outcome

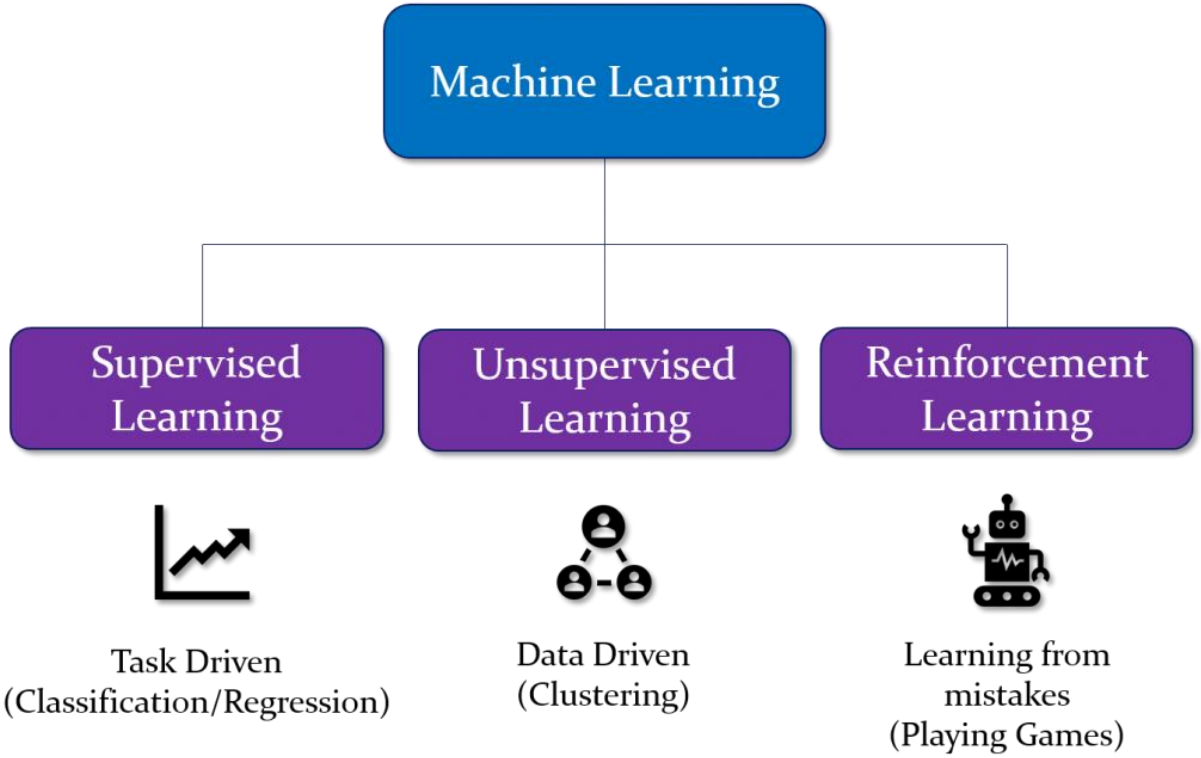


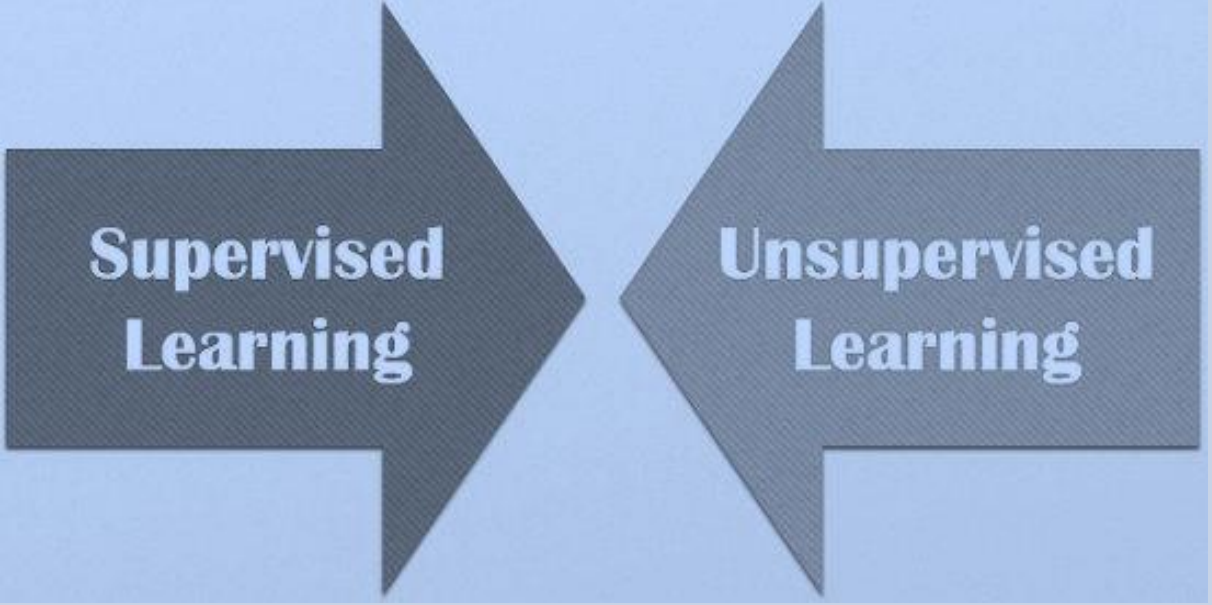
Machine Learning Process

- **Step1** • Define the objective of the Problem Statement
- **Step 2** • Data Gathering
- **Step 3** • Data Preparation
- **Step 4** • Exploratory Data Analysis
- **Step 5** • Building a Machine Learning Model
- **Step 6** • Model Evaluation & Optimization
- **Step 7** • Predictions



Machine Learning Types





Supervised Vs Unsupervised Learning



Learnings

Learning:

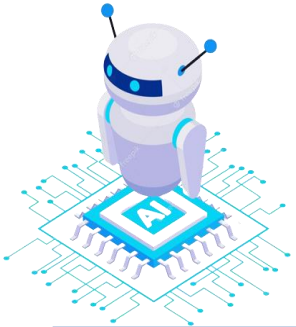
- Building black of AI.
- Method of improving an AI program's knowledge by making observations about its surroundings.

Supervised Learning

Unsupervised Learning

Semi-Supervised Learning

Re-enforcement Learning



Back to main

Supervised Learning

Back to main

Supervised Learning

- Machine learning approach that uses **labelled** datasets.
- Some data tagged with correct answer.
- Helps to predict/forecast outcomes for unexpected data.
- At least some of this approach involves human control. The vast bulk of data is unlabeled and unprocessed.
- Models need to define mapping function to map input variable **(X)** with the output variable **(Y).**
- **Y=f(X)**
- Organizations use to tackle a range of real-world problems, such as spam classification in a distinct folder from your email.

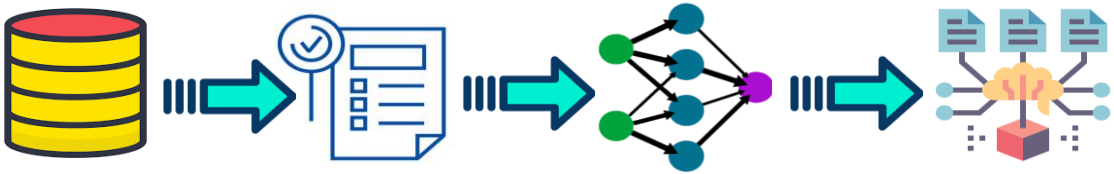
Why Supervised Learning?

- Collect data or create a data output from past experience with supervised learning.
- With the help of experience, it is possible to improve performance criteria.
- Use supervised machine learning to tackle a variety of real-world situations.

How Supervised Learning works?

- each models to produce the desired output.
- training dataset contains both correct and incorrect outputs, allowing the model to improve over time.
- Loss function used to assess the algorithm's correctness and adjusted until the error is suitably minimized.

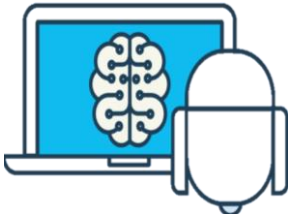
Learning Phase:



Training Data | **Clean, prepare & manipulate data** | **Algorithm** | **Model**

Unsupervised Learning

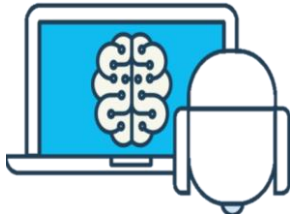
- Machine learning approach that uses **unlabeled** datasets.
- Some data tagged with incorrect answer.
- Helps to predict/forecast outcomes for unexpected data.
- Algorithms discover hidden patterns in data without the need for human intervention.
- suited to answer questions about unseen trends and relationships within data itself.
- Allow you to perform more complex processing tasks.



https://datasciencetoday.net/

Why Unsupervised Learning?

- Finds all kind of unknown patterns in data.
- Find features which can be useful for categorization.
- Taken place in real time, so all input data to be analyzed and labeled in the presence of learners.
- Easier to get unlabeled data from a computer than labeled data, which needs manual intervention.

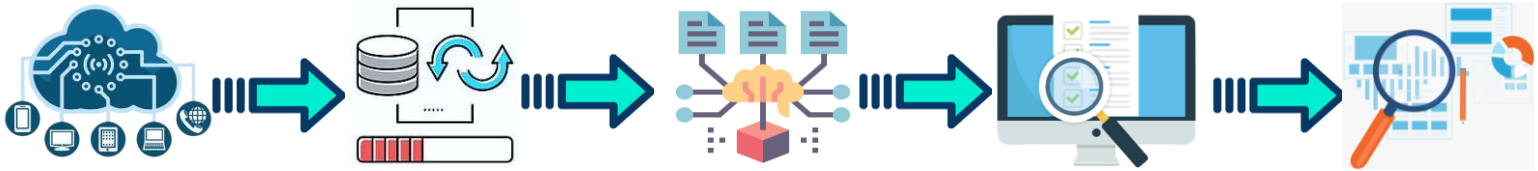


https://datasciencetoday.net/

How Unsupervised Learning works?

- Works by analyzing uncategorized, unlabeled data and finding hidden structures in it.
- Learning process begins with the data scientists training the algorithms using the training datasets.
- Data points in these datasets are unlabeled and uncategorized

Learning Phase:



Get Data

Clean, prepare & manipulate data

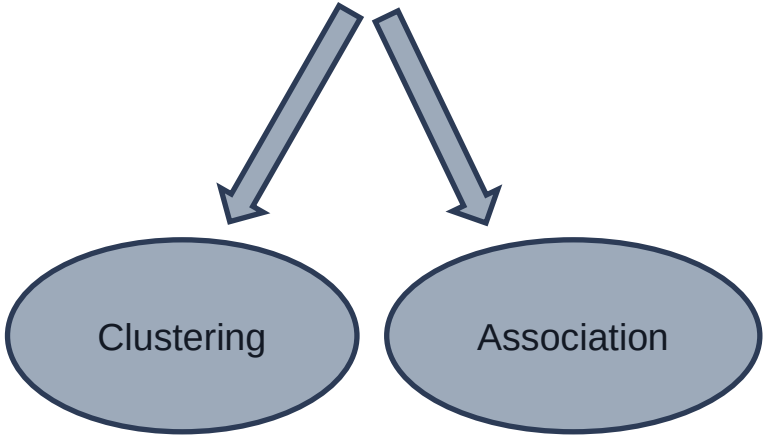
Train Model

Test Data

Improve/ Validate Data

Categories

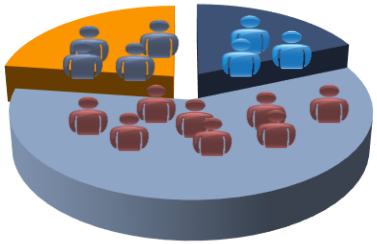
Two types of categories



Clustering

- **Clustering**:
- Data mining technique for grouping unlabeled data based on their similarities or differences.
- Mainly deals with finding a structure or pattern in a collection of uncategorized data.
- process data and find natural clusters(groups) if they exist in the data.
- Modify how many clusters pf algorithms should identify also allows to adjust the granularity of these groups.

Examples:



Market segmentation



https://www.istockphoto.com/

Identify Fake News

Association

❖ **Association**:

- Allow to establish associations amongst data objects inside large databases.
- Discovering exciting relationships between variables in large databases.

Examples:

- people that buy a new home most likely to buy new furniture.
- In market basket analysis and recommendation engines, along the lines of “Customers Who Bought food Item **Pizza** Also Bought” **coke** as a recommendations .



https://blog.vantagecircle.com/

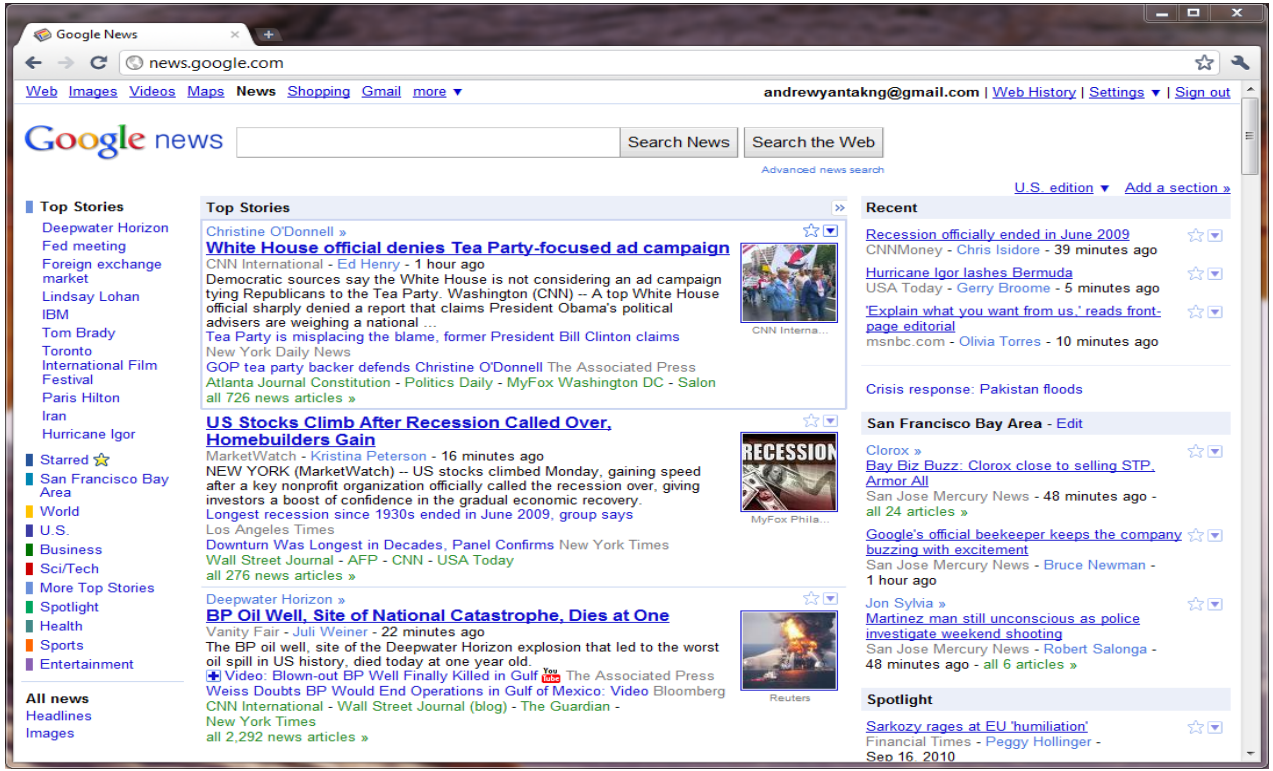
Buy new house also buy new furniture



https://www.redbubble.com/

Pizza coke recommendations

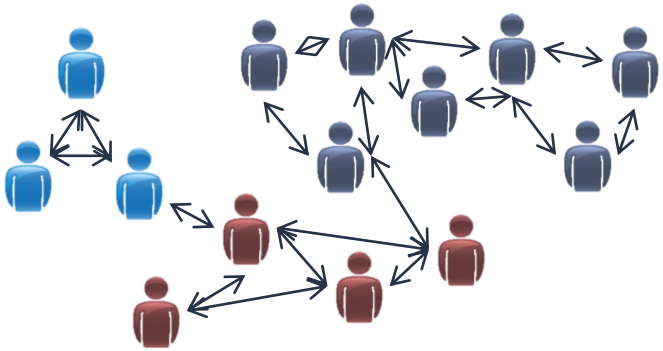
Examples



Google News

- Use different clusters for same data about news articles

Examples



Social network analysis

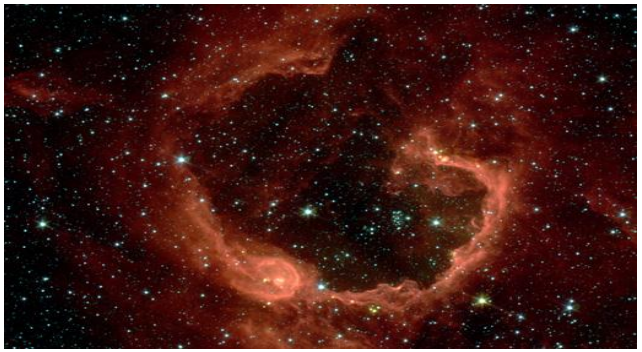


Image credit: NASA/JPL-Caltech/E. Churchwell (Univ. of Wisconsin, Madison)

Astronomical data analysis

Class Task

following examples, which would you address using an unsupervised learning algorithm? (Check all that apply.)

- Given email labeled as spam/not spam, learn a spam filter.
- Given a set of news articles found on the web, group them into set of articles about the same story.
- Given a database of customer data, automatically discover market segments and group customers into different market segments.
- Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

Supervised VS Unsupervised Learning

Parameters	Supervised machine learning technique	Unsupervised machine learning technique
Process	Input and output variables will be given.	only input data will be given
Input Data	Algorithms are trained using labeled data.	Algorithms are used against data which is not labeled
Algorithms Used	Support vector machine, Neural network, Linear and logistics regression, random forest, and Classification trees.	divided into different categories: like Cluster algorithms, K-means, Hierarchical clustering, etc.
Computational Complexity	simpler method.	computationally complex
Accuracy of Results	Highly accurate and trustworthy method.	Less accurate and trustworthy method.
Number of Classes	Number of classes is known.	Number of classes is not known.
Main Drawback	Classifying big data can be a real challenge in Supervised Learning.	cannot get precise information regarding data sorting, and the output as data used in unsupervised learning is labeled and not known.

Back to main

Semi-supervised Learning

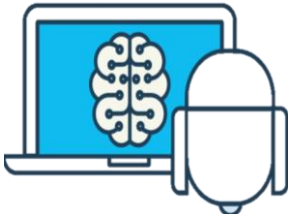
Back to main

Semi-supervised Learning

- combination of both supervised and unsupervised learning.
- Involves small number of labeled examples and large number of unlabeled examples.
- Provides benefits of both supervised and unsupervised learning.
- An ideal for medical images, where small amount of training data can lead to significant improvement in accuracy.
- Helps to predict/forecast outcomes more accurately.

For example,

Radiologist label small subset of CT scans for tumors or diseases so the machine can more accurately predict which patients might require more medical attention.



https://datasciencetoday.net/

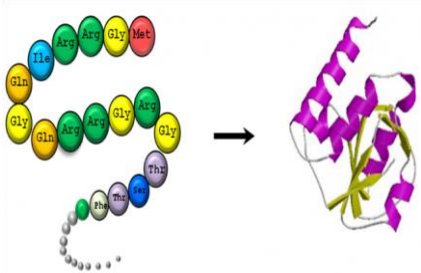
Real-world applications of Semi-supervised Learning

Speech Analysis



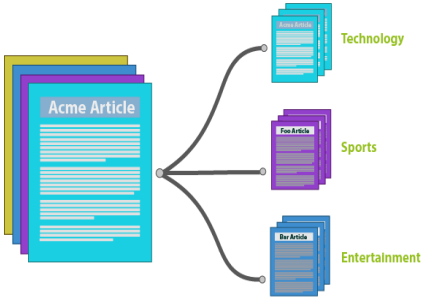
https://www.toolbox.com/

Protein Sequence Classification



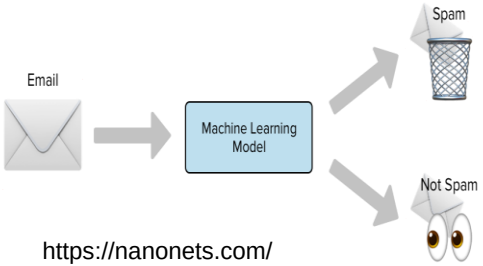
https://www.ebi.ac.uk/

Web Content Classification



https://towardsdatascience.com/

Text Document Classifier

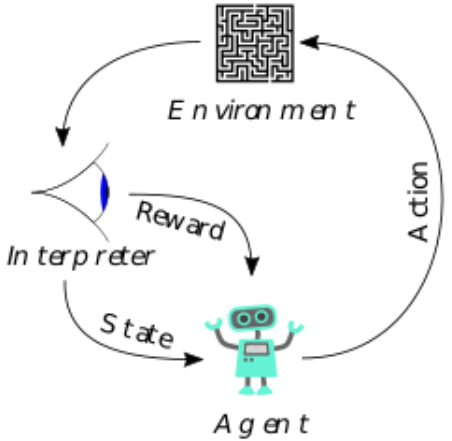


https://nanonets.com/

Back to main

Reinforcement Learning

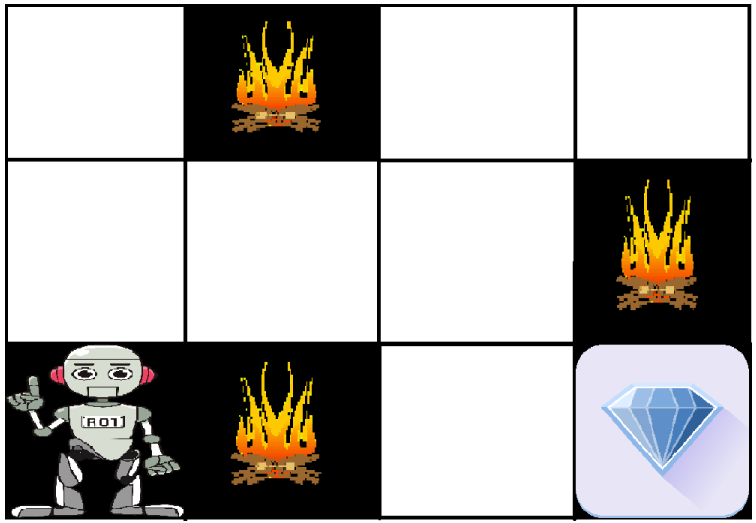
- machine learning training method based on rewarding desired behaviors and/or punishing undesired ones.
- In general, agent is able to perceive and interpret its environment.
- Take actions and learn through trial and error.
- Assigns positive values to desired actions to encourage the agent and negative values to undesired behaviors.
- This method has been adopted in artificial intelligence (AI) as a way of directing unsupervised machine learning through rewards and penalties.
- There is no answer but the reinforcement agent decides what to do to perform the given task.
- In the absence of a training dataset, it is bound to learn from its experience.



https://en.wikipedia.org/

Example:

The problem is as follows: We have an agent and a reward, with many hurdles in between. The agent is supposed to find the best possible path to reach the reward. The following problem explains the problem more easily



https://www.geeksforgeeks.org/